Numerical Methods: Euler's Step by Step Method

Questions 1–18

Question 1. The function $y(x)$ satisfies $\frac{dy}{dx} = e^{-x^2}$ where $y(2) = 1$. For each of the following, use $y_{n+1} = y_n + hf(x_n)$ to approximate $y(3)$ with the given value of h, giving your answer to three decimal places:

a) $h = 0.5$

b) $h = 0.25$

Question 2. The function $y(x)$ satisfies $\frac{dy}{dx} = x^2 - y$ where $y(0) = 1$. For each of the following, use $y_{n+1} = y_n + hf(x_n, y_n)$ to approximate $y(0.4)$ with the given value of h, giving your answer to three decimal places:

a) $h = 0.2$

b) $h = 0.1$

Question 3. The function $y(x)$ satisfies $\dfrac{dy}{dx} = \sin x - y$ where $y(0) = 2$. Calculate $y(0.4)$ using Euler's step by step method for the given value of h, giving your answer to three decimal places:

a) $h = 0.2$

b) $h = 0.1$

Question 4. The function $y(x)$ satisfies the differential equation $\dfrac{dy}{dx} = \dfrac{x^2 + y}{y^2 - x}$ and the condition $y(2) = -1$. Use two iterations of Euler's step by step method to estimate the value of $y(3)$.

Question 5. For each of the following, use Euler's step by step method:

a) $2x\dfrac{dy}{dx} = e^{\sqrt{x}}$ passes through the point $(4, 1)$. Estimate $y(5)$ using $h = 0.5$, giving your answer to three decimal places.

b) $\dfrac{dy}{dx} = \ln(x^2 + y)$ passes through the point $(0, 2)$. Estimate $y(1.5)$ using $h = 0.5$, giving your answer to three decimal places.

Question 6. The function $y(x)$ satisfies $\dfrac{dy}{dx} = 2e^{2x}$ and the condition $y(0) = 1$.

a) Use Euler's step by step method with $h = 0.2$ to approximate $y(0.6)$, giving your answer to three decimal places.

b) Find the exact value of $y(0.6)$.

c) Find the relative error.

d) State whether the Euler approximation to $y(0.6)$ is an over- or an under-estimate of the exact value, giving a reason for your answer.

Question 7. The function $y(x)$ satisfies $\dfrac{dy}{dx} = x + 3\cos y$ and the condition $y(1) = 4$.

a) Use Euler's step by step method with $h = 0.4$ to approximate $y(2.2)$, giving your answer to three decimal places.
b) Comment on the reliability of your result.
c) Explain how you could use this formula to find an improved approximation to $y(2.2)$.

Question 8. A drug concentration C mg depends on how much is already in the bloodstream and the time t hours since infusion started. It may be modelled by $\dfrac{dC}{dt} = 5 - 0.2C + 0.1t$. Initially there is no drug in the bloodstream. Use Euler's method with $h = 0.5$ hours to estimate the concentration after 1 hour.

Question 9. A town's population y of people grows depending on the current population and the season. It may be modelled by $\frac{dy}{dx} = 0.005y + 200\sin\left(\frac{\pi}{6}x\right)$ where x is the number of months passed and is measured in radians. Initially the population is 12,000 people. Use Euler's method with $h = 1$ month to approximate the population after 3 months.

Question 10. The temperature of a rod T°C over time t minutes may be modelled by $\frac{dT}{dt} = 0.2t - 0.05T$. When 5 minutes have passed, the rod is at 50°C. Use Euler's method with $h = 1$ minute to estimate the temperature after 8 minutes, giving your answer to three significant figures.

Question 11. The balance £B on an account grows depending on both balance and a time-varying interest rate, and may be modelled by $\frac{dB}{dt} = 0.01B + 5\cos(0.5t)$ where t months have passed. The account is checked after 2 months and the balance is £970. Use Euler's method twice to estimate the balance after a further 2 months. **Prerequisite Knowledge for Q12: Integration of Standard Functions**

Question 12. The function $y(x)$ passes through $(0, 1)$ and satisfies the differential equation $\frac{dy}{dx} = \sin x$.

a) Use three iterations of Euler's step by step method to estimate $y(0.3)$, giving your answer to three decimal places.

b) Explain why this method underestimates the true value of $y(0.3)$.

Question 13. A curve satisfies $\frac{dy}{dx} = \frac{2x - y}{xy}$ and passes through the point $(3, a)$, where $a > 2$. Using Euler's step by step method, with starting point $(3, a)$, gives an estimate of $y = 4.450$ when $x = 3.2$. Find the value of a.

Question 14. The function $y(x)$ satisfies the differential equation $\frac{dy}{dx} = \sqrt{2x + y^2}$ and the condition $y(1) = 0.8$. After one iteration of Euler's formula, $y_{n+1} = y_n + \frac{1}{k} f(x_n)$, for integer k, it is found that $y_1 = 1.1250$, which was correct to four decimal places.

a) Find the value of $\frac{1}{k}$.

b) Use your value for $\frac{1}{k}$ to calculate (x_2, y_2).

Prerequisite Knowledge for Q15: First Order Linear Differential Equations

Question 15. The curve $y(x)$ satisfies $\dfrac{dy}{dx} = x + y$ where $y(0) = 1$.

a) Explain geometrically what the Euler formula $y_{n+1} = y_n + h(x_n + y_n)$ represents.

b) Use Euler's method with step size $h = 0.5$ to approximate $y(1)$.

c) Find the exact value of $y(1)$.

d) On a sketch, show the curve for $x > 0$, the tangent at $(0, 1)$ and the first Euler step.

Question 16. A company monitors the total amount of data y GB uploaded to its servers during a live event. It is observed that the upload rate is directly proportional to the time t hours since the event began. After 4 hours, the upload rate is measured to be 24 GB per hour.

a) Determine the differential equation to model the total data uploaded.

It is observed that one hour into the event, 1.5 GB has been uploaded.

b) Use Euler's method with $h = 0.25$ to find the amount of data processed after 90 minutes, giving your answer to two decimal places.

Question 17. A post on a social media platform spreads as users share it. At time t hours since posting, the number of people who have seen the post is N. The rate of change of N is affected by:

- sharing at a rate proportional to the current reach, $0.08N$ people per hour
- a reduction in visibility due to platform algorithms at a rate of $500e^{-t}$ people per hour

After 1 hour, the number of people who have seen the post is 4700. Use Euler's method with a step length of 0.5 to estimate the number of people who have seen the post after 2 hours have passed.

Question 18. A leaking container is being monitored. The volume of water W litres changes over time t hours due to three factors:

- Rainfall adds water at an average constant rate of 0.5 litres per hour.
- Evaporation over each day removes water at a rate of $2\sin^2\left(\frac{\pi}{12}t\right)$ litres per hour.
- Leakage removes water proportional to the current volume in the pond, at a rate of $0.01W$ litres per hour.

At the start of observation (at time $t = 0$), the pond contains 100 litres of water. Use Euler's step by step method three times to estimate the amount of water in the container after 9 hours have passed. Give your answer to three significant figures.